

Identifying Variables — Set A — Answer Key

Level 2 · Ages 9-11 · Answers in bold. Accept any answer that means the same.

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Priya lets a toy car roll down a ramp set at heights of 10 cm, 20 cm and 30 cm. She measures how far the car rolls across the floor. She uses the same car and the same ramp every time.

Independent variable: The height of the ramp (10, 20 and 30 cm).

Dependent variable: How far the car rolls across the floor (cm).

Two controlled variables: The same car and the same ramp. (Also accept: the same floor.)

- ② Jack ties a metal washer to a string and lets it swing. He tries strings 20 cm, 40 cm and 60 cm long with the same washer, pulling it back the same distance each time. He counts the swings in 30 seconds.

Independent variable: The length of the string (20, 40 and 60 cm).

Dependent variable: The number of swings in 30 seconds.

Two controlled variables: Any two: the same washer; pulled back the same distance; 30 seconds.

- ③ An adult pours 200 mL of warm water at 45 °C into each of three identical cups. Omar wraps one cup in foil, one in wool and one in newspaper. After 20 minutes he measures the temperature of the water in each cup.

Independent variable: The wrapping material (foil, wool or newspaper).

Dependent variable: Temperature after 20 minutes, °C (or the drop in temperature).

Two controlled variables: Any two: 200 mL of water; starting at 45 °C; identical cups; 20 minutes.